Computer Science & DA

Warehousing

Lecture - 03

By- Vishvadeep Gothi sir

Recap of Previous Lecture

sat, sund ⇒ 10:30 am

mond - friday ⇒ 7:30 am

Topic ETL

Topic ELT Based Warehousing

Topic Data Mart

Topics to be Covered

- Topic: Datawarehouse Schema
- Topic: Star Schema
- Topic: Snowflake Schema

Relational database :-

key no

Accounts

Acno	Custmer Id	Cnam	actype	am	Branchid
A1					B1 ⋮

foreign key

Branch

Branchid	Branch nam
B1	
B2	
B3	
B4	

Topic : Datawarehouse Design Schema

Description, represented by objects such as tables and indexes, of how data relates logically within a data warehouse

Topic : Datawarehouse Schema

1. Star Schema
2. Snowflake Schema
3. Fact Constellation Schema

Topic : Dimension Table

Dimension Tables hold the descriptive information for all related fields that are included in the fact table's records.

⇒ Dimension table holds data of one theme.

Topic : Fact Table

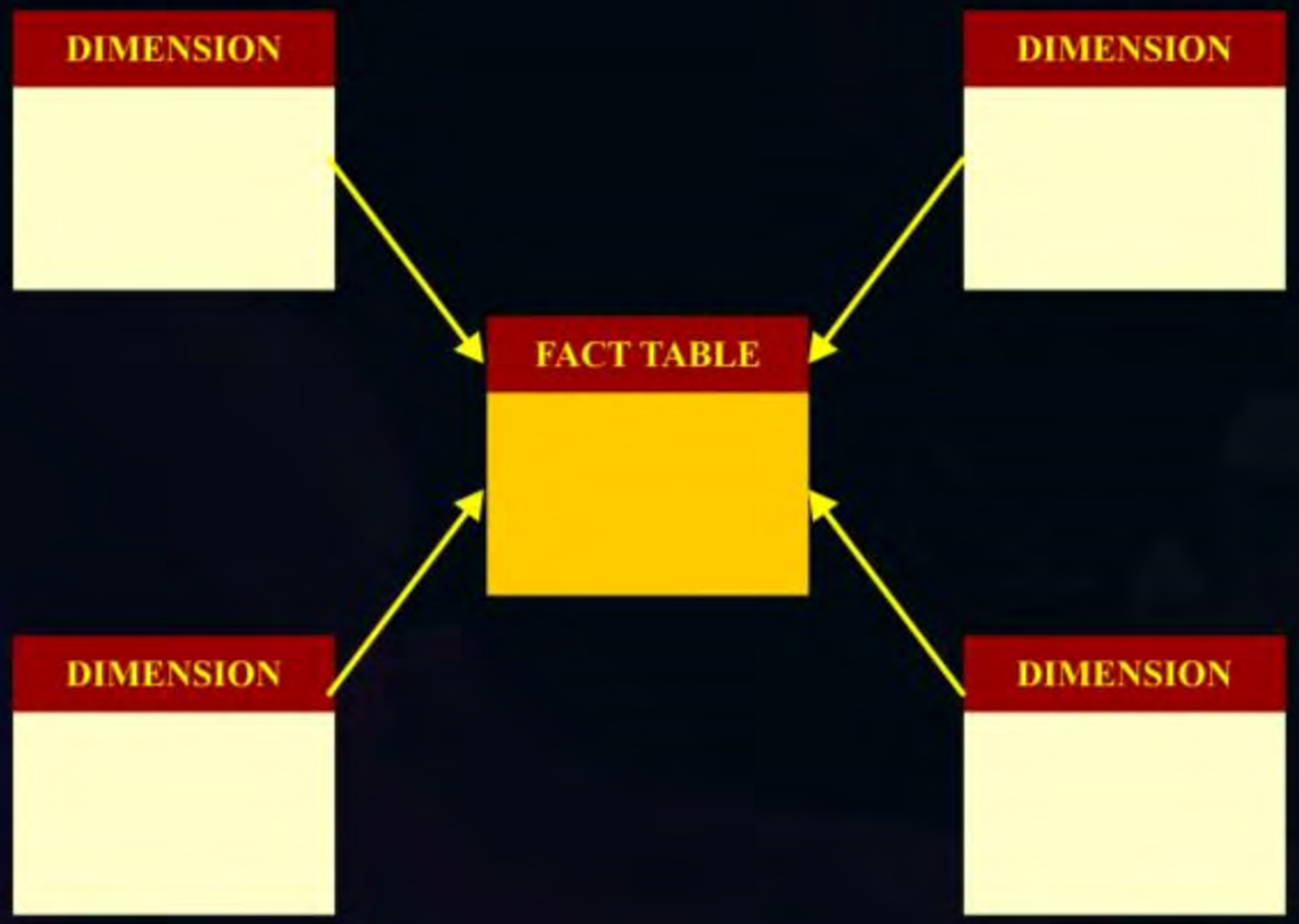
A fact table aggregates metrics, measurements, or facts about business processes

Topic : Fact Table

Topic : Star Schema

Consists of one or more fact tables referencing any number of dimension tables

Topic : Star Schema



Topic : Star Schema

Customer Dimension:

- Customer ID
- Customer Name
- Customer Address
- Customer Email

Product Dimension:

- Product ID Product
- SKU Product
- Category Price

fact

Sales ~~Dimension:~~

- Product ID
- Order ID
- Quantity Ordered
- First Time Customer
- Total

order

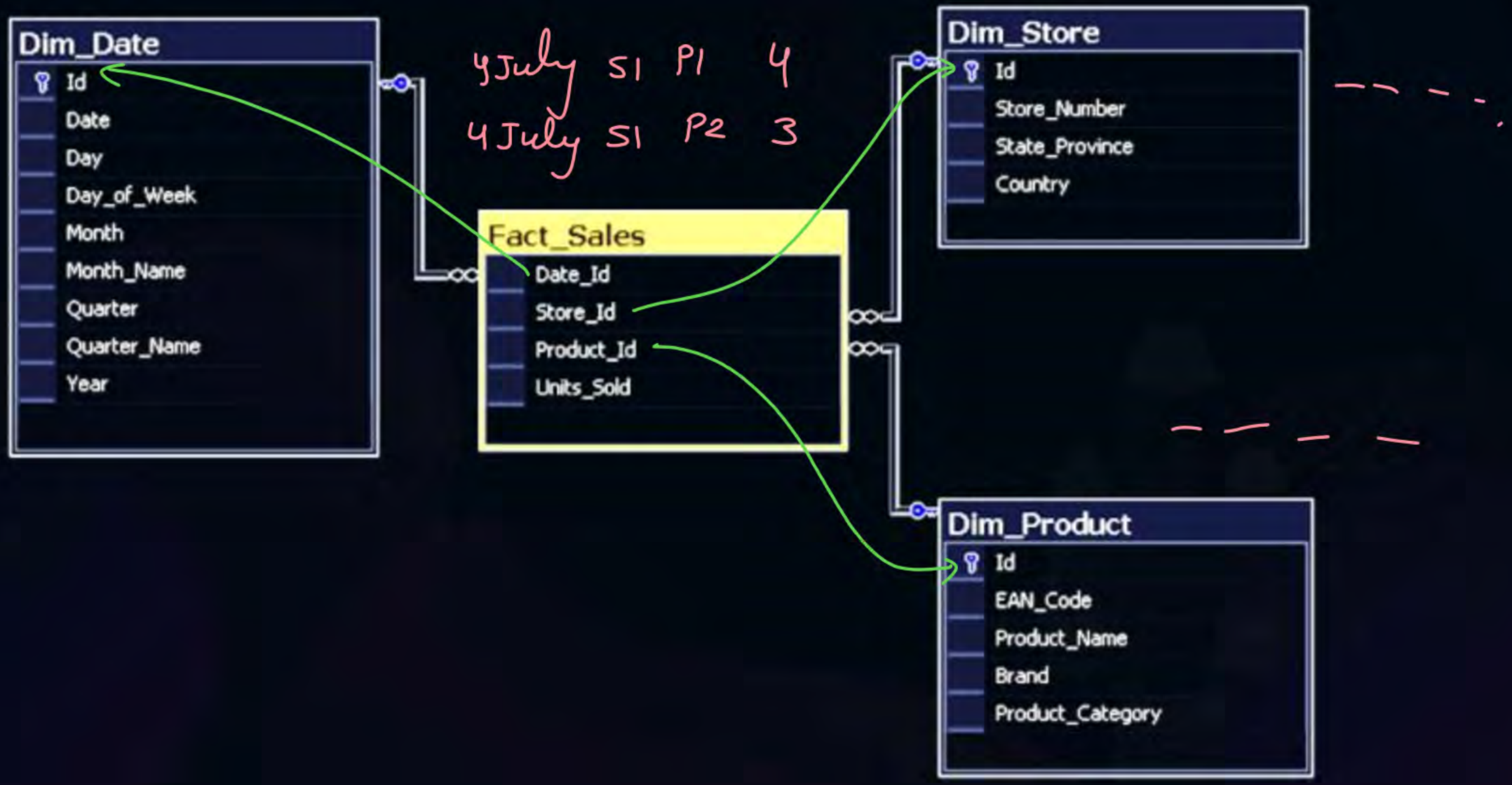
~~Time~~ Dimension:

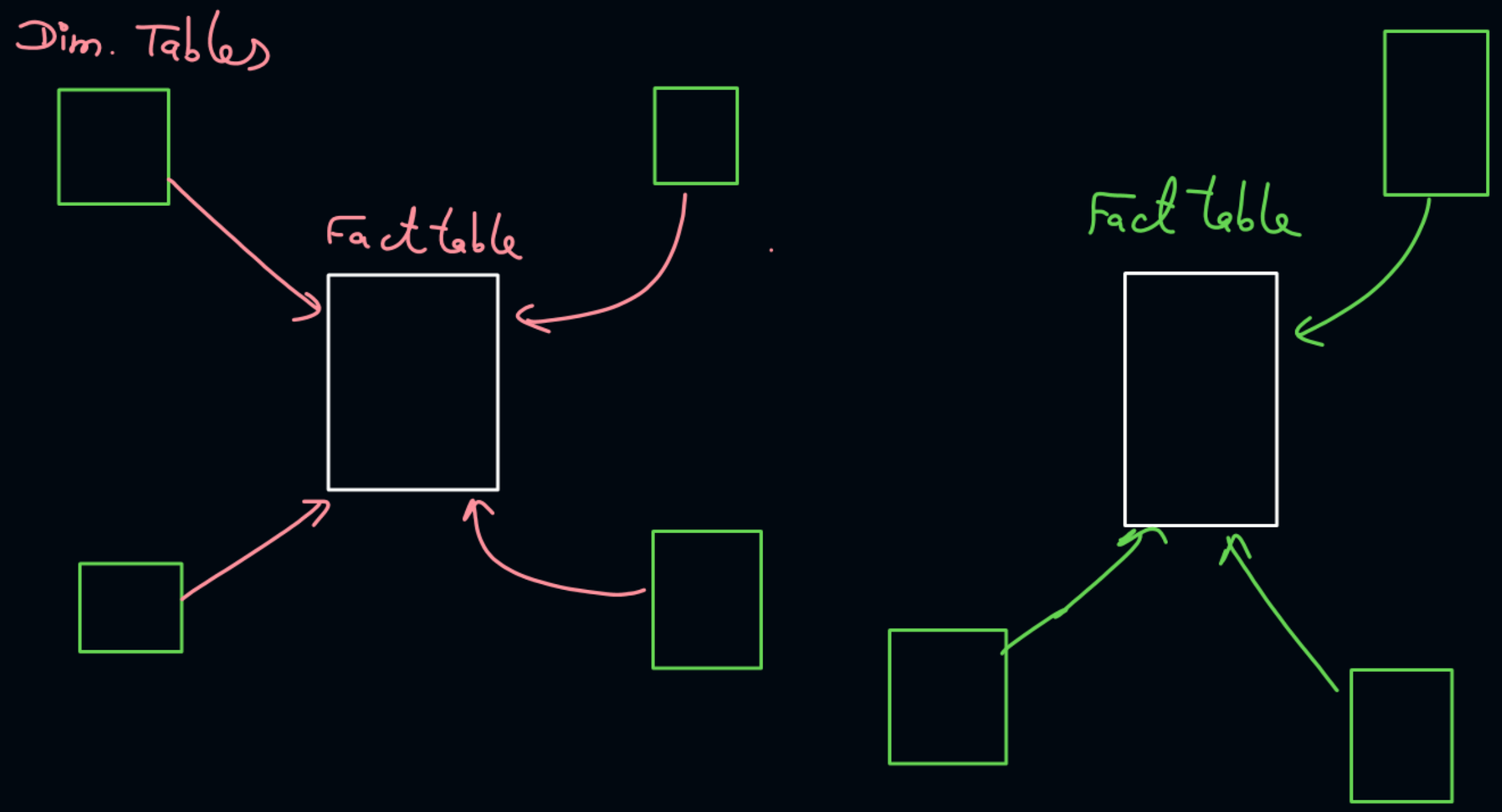
- Order ID
- Order Date
- Year
- Month

Company Dimension:

- Employee ID
- Employee Name
- Department
- Region

Topic : Star Schema




Dim. Tables
Fact table
Fact table

Topic : Star Schema

not normalized

- Star schemas are denormalized
- The primary key in the dimension table is joined to the fact table by the foreign key
- Each dimension in the star schema maps to one dimension table
- Dimension tables within a star scheme are not to be connected directly
- Star schema creates denormalized dimension tables

Topic : Star Schema

- **Benefits:**
  - Simpler queries
  - Simplified business reporting logic
  - Query performance gains (less SQL joins)
  - Fast aggregations
  - Feeding cubes

Topic : Snowflake Schema

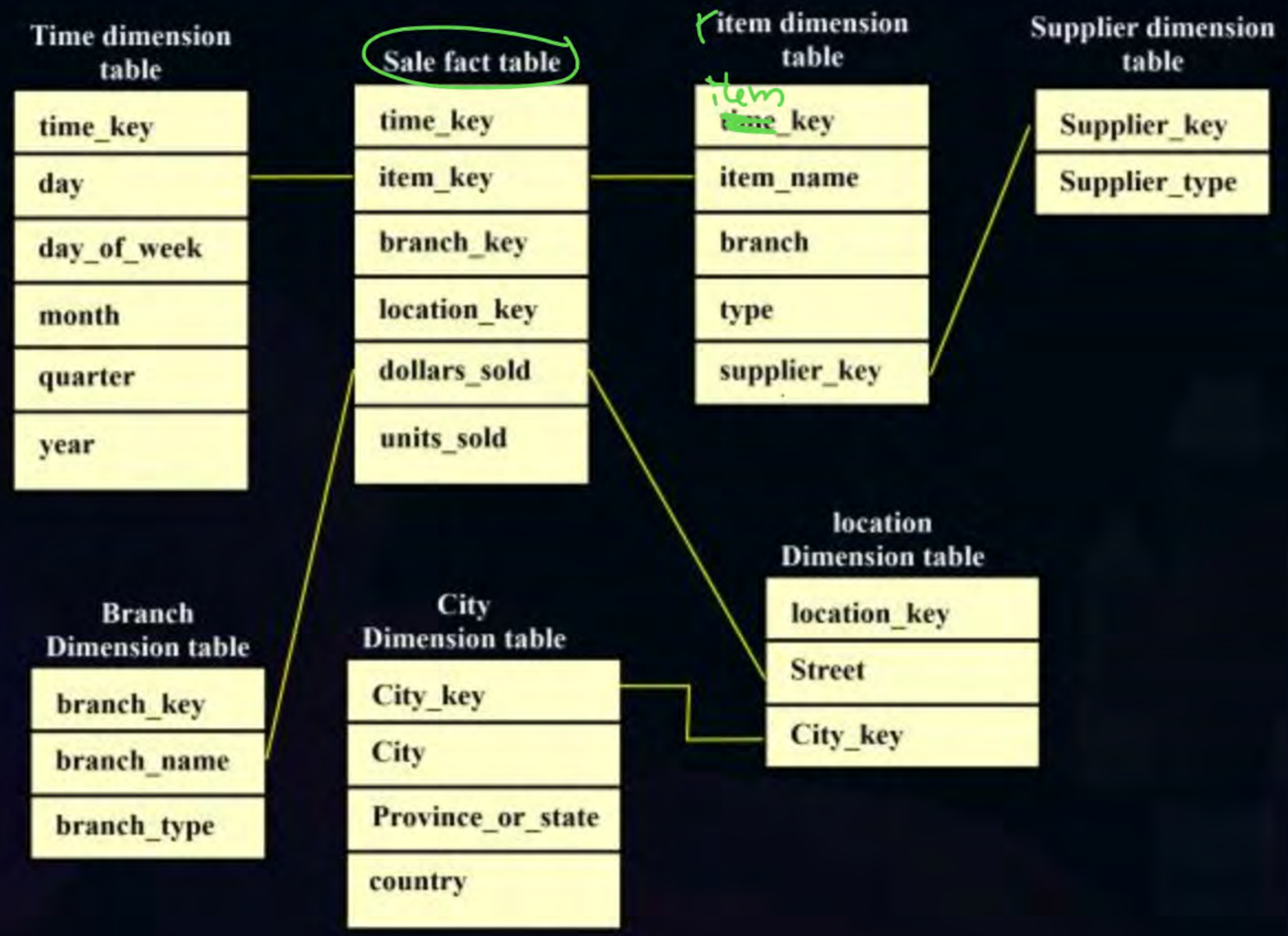
logical arrangement of tables in a multidimensional database such that the entity relationship diagram resembles a snowflake shape

normalize dimension tables of star schem } ⇒ snowflake schema

Topic : Snowflake Schema

Represented by centralized fact tables which are connected to multiple dimensions. "Snowflaking" is a method of normalizing the dimension tables in a star schema

Topic : Snowflake Schema



Topic : Star Schema

Topic : Snowflake Schema

Topic : Snowflake Schema

Advantages:

- Some OLAP multidimensional database modeling tools are optimized for snowflake schemas.
- Normalizing attributes results in storage savings, the tradeoff being additional complexity in source query join

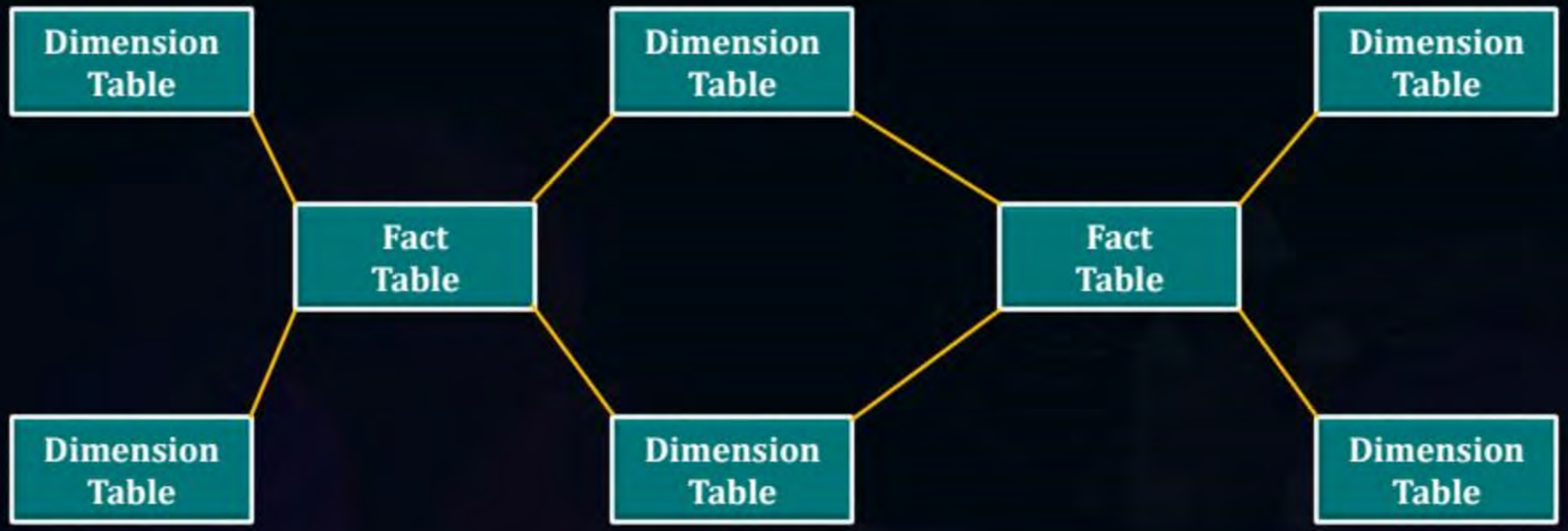
Disadvantages:

- Additional levels of attribute normalization adds complexity to source query joins
- Poor Performance when browsing the joins

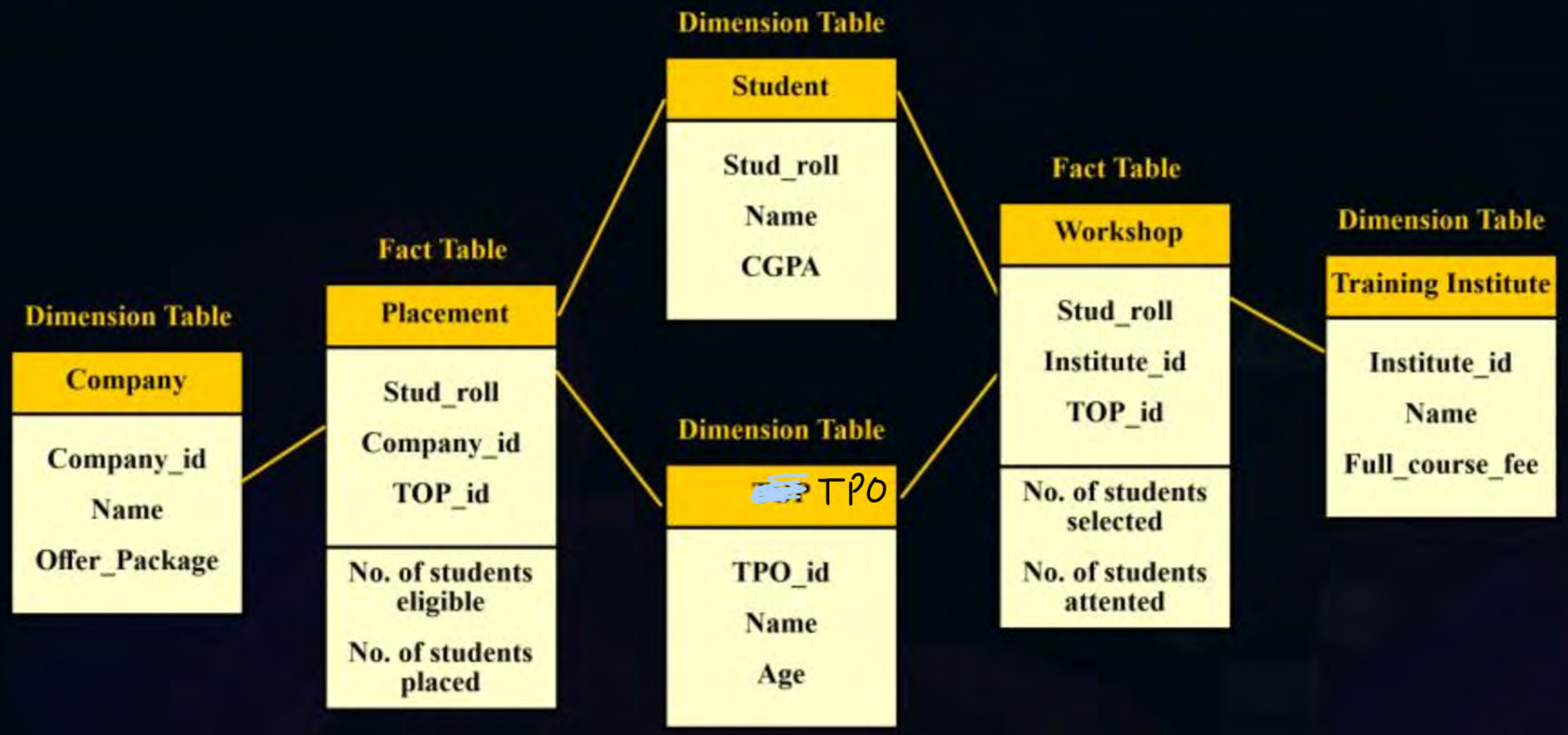
Topic : Fact Constellation Schema (Galaxy Schema)

- Measure of online analytical processing, which is a collection of multiple fact tables sharing dimension tables, viewed as a collection of stars.
- It can be seen as an extension of the star schema.

Topic : Fact Constellation Schema (Galaxy Schema)



Topic : Fact Constellation Schema (Galaxy Schema)



Topic : Fact Constellation Schema (Galaxy Schema)

Advantage:
Provides a flexible schema

Disadvantage:
It is much more complex and hence, hard to implement and maintain

Topic : Fact Constellation Schema (Galaxy Schema)

- Tables are subdivided into fact and dimensional to understand the relationship between them.
- It is a flexible schema that makes users use it.
- Here dimensional tables are shared by the number of fact tables.
- We can access the data in the ~~database~~ warehouse using complex queries.
- It is difficult to understand as it is a very complex schema to implement.
- It uses more space in the database comparative to the star schema.
- It has many joins between dimensional and fact tables and thus it is difficult to understand.
- This is difficult to maintain and operate.

Topic : Data Transformation

Technique used to convert the raw data into a suitable format that efficiently eases data mining and retrieves strategic information

It changes the format, structure, or values of the data and converts them into clean, usable data.

Topic : Data Transformation

1. Data Cleaning
2. Data Smoothing
3. Attribute Construction
4. Data Aggregation
5. Data Discretization
6. Data Generalization
7. Data Normalization

Topic : Data Cleaning

Removing or correcting errors, inconsistencies, and missing values in the data

Employee

Eid	Ename	lastname	phone	add.	phone 2
1					NULL
2					NULL
3					
4					

Topic : Data Smoothing

Process to remove noise from the dataset using some algorithms

It allows for highlighting important features present in the dataset

attributes
or
columns

Topic : Attribute Construction

The new attributes consult the existing attributes to construct a new data set that eases data mining

Example:

Plot area from plot length & width

Length	width	
20 feet	50 feet	⇒ 1000 sqft
25 feet	40 feet	1000 sqft

Property		
.....	Length	width

Property	
- - -	area

Topic : Data Aggregation

The method of storing and presenting data in a summary format

Year 2020

Year 2019

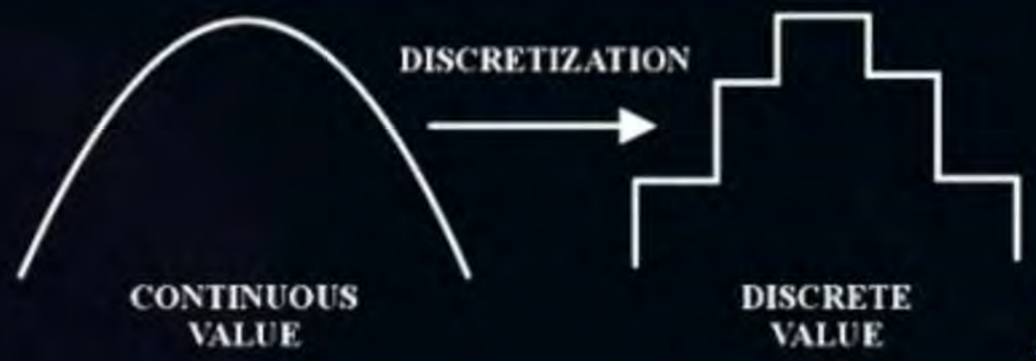
Year 2018

Quater	Cost
Q1	$224,000
Q2	$408,000
Q3	$350,000
Q4	$586,000

Year	Cost
2018	$1,568,000
2019	$2,356,000
2020	$3,594,000

Topic : Data Discretization

A process of converting continuous data attribute values into a finite set of intervals with minimal loss of information.



student	
S1	10
S2	12
S3	13
S4	25
S5	30
S6	19

⇒

S1	10–15
S2	10–15
S3	10–15
S4	20–25
S5	30–35
S6	16–19

2 mins Summary

Topic Datawarehouse Schema

Topic Star Schema

Topic Snowflake Schema

Happy Learning

THANK - YOU